

# Method Metrics

## References

- [jqassistant](#)
- [Neo4j Python Driver](#)

## Effective Method Line Count

### Table 1a - Effective method line count distribution

This table shows the distribution of the effective method line count per artifact. For each artifact the number of methods with effective line count = 1,2,3,... is shown to get an overview of how line counts are distributed over methods.

Only the 15 artifacts with the highest method count and their effective method line count distribution (limited by 40) is shown here. The whole table can be found in the CSV report `Effective_Method_Line_Count_Distribution`.

Have a look below to find out which packages and methods have the highest effective lines of code.

| artifactName       | axon-messaging-4.12.1.jar | axon-server-connector-4.12.1.jar | axon-eventsourcing-4.12.1.jar | axon-modelling-4.12.1.jar | axon-test-4.12.1.jar | axon-configuration-4.12.1.jar | axon-spring-boot-autoconfigure-4.12.1.jar | axon-disruptor-4.12.1.jar | axon-tracing-opentelemetry-4.12.1.jar |
|--------------------|---------------------------|----------------------------------|-------------------------------|---------------------------|----------------------|-------------------------------|-------------------------------------------|---------------------------|---------------------------------------|
| effectiveLineCount |                           |                                  |                               |                           |                      |                               |                                           |                           |                                       |
| 1                  | 2959                      | 585                              | 573                           | 532                       | 279                  | 316                           | 277                                       | 84                        | 16                                    |
| 2                  | 865                       | 205                              | 204                           | 160                       | 172                  | 149                           | 107                                       | 31                        | 4                                     |
| 3                  | 699                       | 141                              | 127                           | 141                       | 66                   | 36                            | 22                                        | 30                        | 10                                    |
| 4                  | 288                       | 69                               | 64                            | 58                        | 49                   | 42                            | 22                                        | 8                         | 7                                     |
| 5                  | 238                       | 48                               | 39                            | 46                        | 24                   | 16                            | 14                                        | 5                         | 2                                     |
| 6                  | 177                       | 27                               | 34                            | 45                        | 18                   | 17                            | 11                                        | 6                         | 3                                     |
| 7                  | 115                       | 19                               | 32                            | 25                        | 20                   | 3                             | 12                                        | 2                         | 2                                     |
| 8                  | 88                        | 22                               | 12                            | 10                        | 11                   | 8                             | 8                                         | 0                         | 0                                     |
| 9                  | 86                        | 13                               | 18                            | 17                        | 10                   | 8                             | 5                                         | 4                         | 1                                     |
| 10                 | 50                        | 15                               | 7                             | 8                         | 4                    | 6                             | 4                                         | 3                         | 0                                     |
| 11                 | 46                        | 12                               | 2                             | 7                         | 9                    | 5                             | 3                                         | 1                         | 0                                     |
| 12                 | 46                        | 12                               | 5                             | 2                         | 9                    | 1                             | 0                                         | 1                         | 1                                     |
| 13                 | 34                        | 7                                | 2                             | 8                         | 4                    | 2                             | 0                                         | 1                         | 0                                     |
| 14                 | 15                        | 1                                | 3                             | 4                         | 2                    | 1                             | 0                                         | 2                         | 0                                     |
| 15                 | 13                        | 5                                | 2                             | 5                         | 3                    | 0                             | 0                                         | 0                         | 0                                     |
| 16                 | 10                        | 3                                | 6                             | 2                         | 4                    | 0                             | 0                                         | 0                         | 0                                     |
| 17                 | 11                        | 7                                | 0                             | 1                         | 4                    | 1                             | 1                                         | 1                         | 0                                     |
| 18                 | 10                        | 0                                | 3                             | 0                         | 3                    | 0                             | 0                                         | 1                         | 0                                     |
| 19                 | 8                         | 4                                | 0                             | 1                         | 0                    | 1                             | 1                                         | 1                         | 0                                     |
| 20                 | 5                         | 2                                | 2                             | 2                         | 1                    | 1                             | 0                                         | 0                         | 0                                     |
| 21                 | 7                         | 1                                | 2                             | 2                         | 1                    | 1                             | 0                                         | 0                         | 0                                     |
| 22                 | 2                         | 0                                | 1                             | 2                         | 1                    | 0                             | 2                                         | 1                         | 0                                     |
| 23                 | 4                         | 1                                | 2                             | 0                         | 1                    | 1                             | 1                                         | 0                         | 0                                     |
| 24                 | 5                         | 3                                | 1                             | 1                         | 1                    | 0                             | 0                                         | 1                         | 0                                     |
| 25                 | 1                         | 0                                | 1                             | 0                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 26                 | 4                         | 0                                | 0                             | 1                         | 1                    | 1                             | 0                                         | 0                         | 0                                     |
| 27                 | 0                         | 1                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 28                 | 2                         | 2                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 29                 | 0                         | 0                                | 0                             | 0                         | 1                    | 0                             | 0                                         | 0                         | 0                                     |
| 30                 | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 31                 | 1                         | 0                                | 0                             | 0                         | 1                    | 0                             | 0                                         | 0                         | 0                                     |
| 32                 | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 1                         | 0                                     |
| 33                 | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 34                 | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 36                 | 2                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 38                 | 0                         | 1                                | 0                             | 1                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 39                 | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 41                 | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 42                 | 0                         | 1                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         | 0                                     |
| 43                 | 0                         | 0                                | 0                             | 0                         | 0                    | 1                             | 0                                         | 0                         | 0                                     |

Table 1b - Effective method line count distribution (normalized)

The table shown here only includes the first 40 rows which typically represents the most significant entries. Have a look below to find out which packages and methods have the highest effective lines of code.

| artifactName       | axon-messaging-4.12.1.jar | axon-server-connector-4.12.1.jar | axon-eventsourcing-4.12.1.jar | axon-modelling-4.12.1.jar | axon-test-4.12.1.jar | axon-configuration-4.12.1.jar | axon-spring-boot-autoconfigure-4.12.1.jar | axon-disruptor-4.12.1.jar | axon-tracing-opentelemetry-4.12.1.jar |
|--------------------|---------------------------|----------------------------------|-------------------------------|---------------------------|----------------------|-------------------------------|-------------------------------------------|---------------------------|---------------------------------------|
| effectiveLineCount |                           |                                  |                               |                           |                      |                               |                                           |                           |                                       |
| 1                  | 50.999655                 | 48.467274                        | 50.175131                     | 49.213691                 | 39.857143            | 51.132686                     | 56.415479                                 | 45.652174                 | 34.782609                             |
| 2                  | 14.908652                 | 16.984258                        | 17.863398                     | 14.801110                 | 24.571429            | 24.110032                     | 21.792261                                 | 16.847826                 | 8.695652                              |
| 3                  | 12.047570                 | 11.681856                        | 11.120841                     | 13.043478                 | 9.428571             | 5.825243                      | 4.480652                                  | 16.304348                 | 21.739130                             |
| 4                  | 4.963806                  | 5.716653                         | 5.604203                      | 5.365402                  | 7.000000             | 6.796117                      | 4.480652                                  | 4.347826                  | 15.217391                             |
| 5                  | 4.102034                  | 3.976802                         | 3.415061                      | 4.255319                  | 3.428571             | 2.588997                      | 2.851324                                  | 2.717391                  | 4.347826                              |
| 6                  | 3.050672                  | 2.236951                         | 2.977233                      | 4.162812                  | 2.571429             | 2.750809                      | 2.240326                                  | 3.260870                  | 6.521739                              |
| 7                  | 1.982075                  | 1.574151                         | 2.802102                      | 2.312673                  | 2.857143             | 0.485437                      | 2.443992                                  | 1.086957                  | 4.347826                              |
| 8                  | 1.516718                  | 1.822701                         | 1.050788                      | 0.925069                  | 1.571429             | 1.294498                      | 1.629328                                  | 0.000000                  | 0.000000                              |
| 9                  | 1.482248                  | 1.077051                         | 1.576182                      | 1.572618                  | 1.428571             | 1.294498                      | 1.018330                                  | 2.173913                  | 2.173913                              |
| 10                 | 0.861772                  | 1.242751                         | 0.612960                      | 0.740056                  | 0.571429             | 0.970874                      | 0.814664                                  | 1.630435                  | 0.000000                              |
| 11                 | 0.792830                  | 0.994200                         | 0.175131                      | 0.647549                  | 1.285714             | 0.809061                      | 0.610998                                  | 0.543478                  | 0.000000                              |
| 12                 | 0.792830                  | 0.994200                         | 0.437828                      | 0.185014                  | 1.285714             | 0.161812                      | 0.000000                                  | 0.543478                  | 2.173913                              |
| 13                 | 0.586005                  | 0.579950                         | 0.175131                      | 0.740056                  | 0.571429             | 0.323625                      | 0.000000                                  | 0.543478                  | 0.000000                              |
| 14                 | 0.258532                  | 0.082850                         | 0.262697                      | 0.370028                  | 0.285714             | 0.161812                      | 0.000000                                  | 1.086957                  | 0.000000                              |
| 15                 | 0.224061                  | 0.414250                         | 0.175131                      | 0.462535                  | 0.428571             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 16                 | 0.172354                  | 0.248550                         | 0.525394                      | 0.185014                  | 0.571429             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 17                 | 0.189590                  | 0.579950                         | 0.000000                      | 0.092507                  | 0.571429             | 0.161812                      | 0.203666                                  | 0.543478                  | 0.000000                              |
| 18                 | 0.172354                  | 0.000000                         | 0.262697                      | 0.000000                  | 0.428571             | 0.000000                      | 0.000000                                  | 0.543478                  | 0.000000                              |
| 19                 | 0.137883                  | 0.331400                         | 0.000000                      | 0.092507                  | 0.000000             | 0.161812                      | 0.203666                                  | 0.543478                  | 0.000000                              |
| 20                 | 0.086177                  | 0.165700                         | 0.175131                      | 0.185014                  | 0.142857             | 0.161812                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 21                 | 0.120648                  | 0.082850                         | 0.175131                      | 0.185014                  | 0.142857             | 0.161812                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 22                 | 0.034471                  | 0.000000                         | 0.087566                      | 0.185014                  | 0.142857             | 0.000000                      | 0.407332                                  | 0.543478                  | 0.000000                              |
| 23                 | 0.068942                  | 0.082850                         | 0.175131                      | 0.000000                  | 0.142857             | 0.161812                      | 0.203666                                  | 0.000000                  | 0.000000                              |
| 24                 | 0.086177                  | 0.248550                         | 0.087566                      | 0.092507                  | 0.142857             | 0.000000                      | 0.000000                                  | 0.543478                  | 0.000000                              |
| 25                 | 0.017235                  | 0.000000                         | 0.087566                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 26                 | 0.068942                  | 0.000000                         | 0.000000                      | 0.092507                  | 0.142857             | 0.161812                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 27                 | 0.000000                  | 0.082850                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 28                 | 0.034471                  | 0.165700                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 29                 | 0.000000                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.142857             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 30                 | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 31                 | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.142857             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 32                 | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.543478                  | 0.000000                              |
| 33                 | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 34                 | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 36                 | 0.034471                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 38                 | 0.000000                  | 0.082850                         | 0.000000                      | 0.092507                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 39                 | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 41                 | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 42                 | 0.000000                  | 0.082850                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.000000                              |
| 43                 | 0.000000                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.161812                      | 0.000000                                  | 0.000000                  | 0.000000                              |

Table 1b Chart 1 - Effective method line count distribution (normalized)

<Figure size 640x480 with 0 Axes>

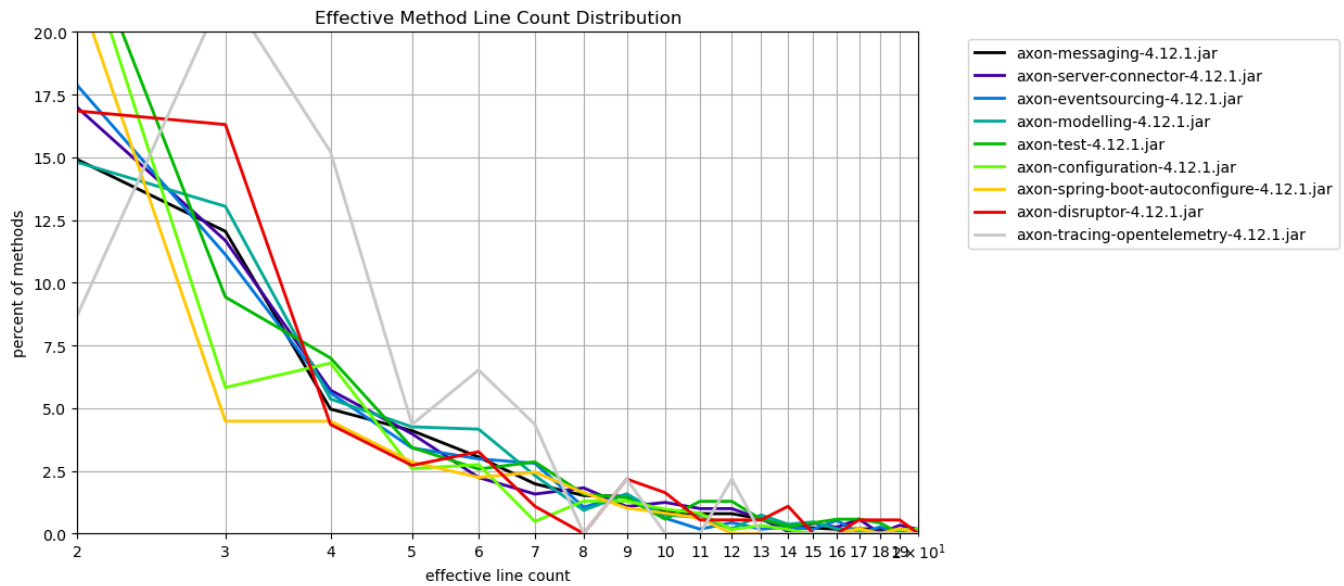


Table 1c - Top 30 packages with highest effective line counts

The following table shows the top 30 packages with the highest effective lines of code. The whole table can be found in the CSV report

Effective\_lines\_of\_method\_code\_per\_package .

|    | artifactName                          | fullPackageName                                   | linesInPackage | methodCount | maxLinesMethod | maxLinesClass                  |
|----|---------------------------------------|---------------------------------------------------|----------------|-------------|----------------|--------------------------------|
| 0  | axon-messaging-4.12.1                 | org.axonframework.eventhandling                   | 2331           | 838         | 64             |                                |
| 1  | axon-configuration-4.12.1             | org.axonframework.config                          | 1615           | 618         | 50             |                                |
| 2  | axon-messaging-4.12.1                 | org.axonframework.eventhandling.pooled            | 1083           | 350         | 81             |                                |
| 3  | axon-test-4.12.1                      | org.axonframework.test.aggregate                  | 956            | 251         | 45             | ap                             |
| 4  | axon-messaging-4.12.1                 | org.axonframework.queryhandling                   | 953            | 388         | 36             |                                |
| 5  | axon-server-connector-4.12.1          | org.axonframework.axonserver.connector.event.axon | 885            | 283         | 28             |                                |
| 6  | axon-messaging-4.12.1                 | org.axonframework.eventhandling.deadletter.jdbc   | 848            | 249         | 31             |                                |
| 7  | axon-modelling-4.12.1                 | org.axonframework.modelling.command               | 848            | 336         | 21             | lambda in                      |
| 8  | axon-server-connector-4.12.1          | org.axonframework.axonserver.connector            | 836            | 335         | 42             |                                |
| 9  | axon-server-connector-4.12.1          | org.axonframework.axonserver.connector.query      | 749            | 219         | 27             |                                |
| 10 | axon-eventsourcing-4.12.1             | org.axonframework.eventsourcing.eventstore        | 712            | 264         | 21             |                                |
| 11 | axon-spring-boot-autoconfigure-4.12.1 | org.axonframework.springboot.autoconfigure        | 694            | 249         | 48             |                                |
| 12 | axon-messaging-4.12.1                 | org.axonframework.messaging.annotation            | 668            | 238         | 23             |                                |
| 13 | axon-modelling-4.12.1                 | org.axonframework.modelling.command.inspection    | 627            | 215         | 26             | inspe                          |
| 14 | axon-eventsourcing-4.12.1             | org.axonframework.eventsourcing                   | 622            | 251         | 20             | c                              |
| 15 | axon-disruptor-4.12.1                 | org.axonframework.disruptor.commandhandling       | 605            | 184         | 32             |                                |
| 16 | axon-eventsourcing-4.12.1             | org.axonframework.eventsourcing.eventstore.leg... | 573            | 187         | 25             |                                |
| 17 | axon-eventsourcing-4.12.1             | org.axonframework.eventsourcing.eventstore.jdbc   | 568            | 236         | 24             |                                |
| 18 | axon-messaging-4.12.1                 | org.axonframework.serialization                   | 548            | 184         | 24             |                                |
| 19 | axon-messaging-4.12.1                 | org.axonframework.eventhandling.deadletter.jpa    | 525            | 132         | 28             |                                |
| 20 | axon-messaging-4.12.1                 | org.axonframework.common                          | 498            | 144         | 24             | getExactDirectSuperTypesOfPara |
| 21 | axon-test-4.12.1                      | org.axonframework.test.saga                       | 495            | 169         | 29             |                                |
| 22 | axon-modelling-4.12.1                 | org.axonframework.modelling.saga                  | 490            | 194         | 22             |                                |
| 23 | axon-messaging-4.12.1                 | org.axonframework.commandhandling.gateway         | 488            | 174         | 50             |                                |
| 24 | axon-messaging-4.12.1                 | org.axonframework.commandhandling.distributed     | 477            | 175         | 23             |                                |
| 25 | axon-messaging-4.12.1                 | org.axonframework.eventhandling.tokenstore.jdbc   | 430            | 130         | 26             |                                |

|    | artifactName          | fullPackageName                                   | linesInPackage | methodCount | maxLinesMethod | maxLinesMethod |
|----|-----------------------|---------------------------------------------------|----------------|-------------|----------------|----------------|
| 26 | axon-messaging-4.12.1 | org.axonframework.eventhandling.deadletter.leg... | 401            | 97          | 21             |                |
| 27 | axon-messaging-4.12.1 | org.axonframework.commandhandling                 | 377            | 169         | 13             |                |
| 28 | axon-modelling-4.12.1 | org.axonframework.modelling.saga.repository.jdbc  | 374            | 84          | 38             |                |
| 29 | axon-messaging-4.12.1 | org.axonframework.messaging.unitofwork            | 363            | 129         | 32             |                |

## Table 1d - Top 30 methods with the highest effective line count

The following table shows the top 30 methods with the highest effective lines of code. The whole table can be found in the CSV report

Effective\_lines\_of\_method\_code\_per\_package .

| index | artifactName                                | fullPackageName                                   | maxLinesMethodType                                       |
|-------|---------------------------------------------|---------------------------------------------------|----------------------------------------------------------|
| 0     | 2<br>axon-messaging-4.12.1                  | org.axonframework.eventhandling.pooled            | Coordinator\$CoordinationTask                            |
| 1     | 0<br>axon-messaging-4.12.1                  | org.axonframework.eventhandling                   | TrackingEventProcessor                                   |
| 2     | 23<br>axon-messaging-4.12.1                 | org.axonframework.commandhandling.gateway         | CommandGatewayFactory                                    |
| 3     | 1<br>axon-configuration-4.12.1              | org.axonframework.config                          | DefaultConfigurer                                        |
| 4     | 11<br>axon-spring-boot-autoconfigure-4.12.1 | org.axonframework.springboot.autoconfig           | AxonAutoConfiguration                                    |
| 5     | 3<br>axon-test-4.12.1                       | org.axonframework.test.aggregate                  | Reporter                                                 |
| 6     | 43<br>axon-messaging-4.12.1                 | org.axonframework.deadline.quartz                 | DeadlineJob                                              |
| 7     | 8<br>axon-server-connector-4.12.1           | org.axonframework.axonserver.connector            | AxonServerConnectionManager\$Builder                     |
| 8     | 61<br>axon-messaging-4.12.1                 | org.axonframework.updates.detection               | TestEnvironmentDetector                                  |
| 9     | 56<br>axon-messaging-4.12.1                 | org.axonframework.updates                         | UpdateCheckerHttpClient                                  |
| 10    | 28<br>axon-modelling-4.12.1                 | org.axonframework.modelling.saga.repository.jdbc  | JdbcSagaStore                                            |
| 11    | 4<br>axon-messaging-4.12.1                  | org.axonframework.queryhandling                   | SimpleQueryBus                                           |
| 12    | 36<br>axon-messaging-4.12.1                 | org.axonframework.messaging.deadletter            | InMemorySequencedDeadLetterQueue                         |
| 13    | 29<br>axon-messaging-4.12.1                 | org.axonframework.messaging.unitofwork            | BatchingUnitOfWork                                       |
| 14    | 15<br>axon-disruptor-4.12.1                 | org.axonframework.disruptor.commandhandling       | DisruptorCommandBus                                      |
| 15    | 6<br>axon-messaging-4.12.1                  | org.axonframework.eventhandling.deadletter.jdbc   | DefaultDeadLetterJdbcConverter                           |
| 16    | 21<br>axon-test-4.12.1                      | org.axonframework.test.saga                       | SagaTestFixture                                          |
| 17    | 19<br>axon-messaging-4.12.1                 | org.axonframework.eventhandling.deadletter.jpa    | DeadLetterEventEntry                                     |
| 18    | 5<br>axon-server-connector-4.12.1           | org.axonframework.axonserver.connector.event.axon | PersistentStreamConnection <i>SegmentConnection</i> P... |
| 19    | 9<br>axon-server-connector-4.12.1           | org.axonframework.axonserver.connector.query      | AxonServerQueryBus                                       |
| 20    | 13<br>axon-modelling-4.12.1                 | org.axonframework.modelling.command.inspection    | AnnotatedAggregateMetaModelFactory\$AnnotatedAg...       |
| 21    | 25<br>axon-messaging-4.12.1                 | org.axonframework.eventhandling.tokenstore.jdbc   | JdbcTokenStore                                           |
| 22    | 16<br>axon-eventsourcing-4.12.1             | org.axonframework.eventsourcing.eventstore.leg... | JpaEventStorageEngine                                    |
| 23    | 42<br>axon-messaging-4.12.1                 | org.axonframework.deadline                        | SimpleDeadlineManager\$DeadlineTask                      |
| 24    | 17<br>axon-eventsourcing-4.12.1             | org.axonframework.eventsourcing.eventstore.jdbc   | JdbcEventStorageEngine                                   |
| 25    | 18<br>axon-messaging-4.12.1                 | org.axonframework.serialization                   | AbstractXStreamSerializer                                |

|    | index | artifactName                 | fullPackageName                                   | maxLinesMethodType                          |
|----|-------|------------------------------|---------------------------------------------------|---------------------------------------------|
| 26 | 20    | axon-messaging-4.12.1        | org.axonframework.common                          | TypeReflectionUtils getExactDirectSuperType |
| 27 | 37    | axon-server-connector-4.12.1 | org.axonframework.axonserver.connector.command    | AxonServerCommandBus                        |
| 28 | 70    | axon-server-connector-4.12.1 | org.axonframework.axonserver.connector.event.util | EventCipher                                 |
| 29 | 58    | axon-messaging-4.12.1        | org.axonframework.deadline.jobrunr                | JobRunrDeadlineManager                      |

## Cyclomatic Complexity

### Table 2a - Cyclomatic method complexity distribution

This table shows the distribution of the cyclomatic complexity of methods per artifact. For each artifact the number of methods with the cyclomatic complexity = 1,2,3,... is shown to get an overview of how cyclomatic complexity is distributed over methods.

Only the 15 artifacts with the highest method count sum and their cyclomatic method complexity distribution (limited by 40) is shown here. The whole table can be found in the CSV report [Cyclomatic\\_Method\\_Complexity\\_Distribution](#).

Have a look below to find out which packages and methods have the highest effective lines of code.

| artifactName         | axon-messaging-4.12.1.jar | axon-server-connector-4.12.1.jar | axon-eventsourcing-4.12.1.jar | axon-modelling-4.12.1.jar | axon-test-4.12.1.jar | axon-configuration-4.12.1.jar | axon-spring-boot-autoconfigure-4.12.1.jar | axon-disruptor-4.12.1.jar | axon-tracing-opentelemetry-4.12.1.jar |
|----------------------|---------------------------|----------------------------------|-------------------------------|---------------------------|----------------------|-------------------------------|-------------------------------------------|---------------------------|---------------------------------------|
| cyclomaticComplexity |                           |                                  |                               |                           |                      |                               |                                           |                           |                                       |
| 1                    | 4648                      | 1012                             | 943                           | 896                       | 521                  | 553                           | 466                                       | 146                       | 3                                     |
| 2                    | 486                       | 91                               | 94                            | 75                        | 61                   | 37                            | 12                                        | 20                        |                                       |
| 3                    | 299                       | 44                               | 53                            | 41                        | 59                   | 18                            | 6                                         | 5                         |                                       |
| 4                    | 154                       | 26                               | 24                            | 29                        | 23                   | 5                             | 3                                         | 4                         |                                       |
| 5                    | 84                        | 8                                | 9                             | 24                        | 13                   | 3                             | 1                                         | 3                         |                                       |
| 6                    | 54                        | 5                                | 4                             | 10                        | 9                    | 0                             | 1                                         | 2                         |                                       |
| 7                    | 23                        | 7                                | 7                             | 2                         | 4                    | 2                             | 1                                         | 2                         |                                       |
| 8                    | 14                        | 6                                | 7                             | 1                         | 2                    | 0                             | 0                                         | 2                         |                                       |
| 9                    | 9                         | 4                                | 0                             | 2                         | 2                    | 0                             | 0                                         | 0                         |                                       |
| 10                   | 6                         | 2                                | 0                             | 0                         | 1                    | 0                             | 0                                         | 0                         |                                       |
| 11                   | 10                        | 0                                | 0                             | 0                         | 2                    | 0                             | 0                                         | 0                         |                                       |
| 12                   | 5                         | 1                                | 0                             | 0                         | 2                    | 0                             | 0                                         | 0                         |                                       |
| 13                   | 2                         | 0                                | 1                             | 0                         | 1                    | 0                             | 0                                         | 0                         |                                       |
| 14                   | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 1                                         | 0                         |                                       |
| 15                   | 1                         | 1                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         |                                       |
| 16                   | 0                         | 0                                | 0                             | 1                         | 0                    | 0                             | 0                                         | 0                         |                                       |
| 17                   | 2                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         |                                       |
| 21                   | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         |                                       |
| 23                   | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         |                                       |
| 40                   | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         |                                       |
| 41                   | 1                         | 0                                | 0                             | 0                         | 0                    | 0                             | 0                                         | 0                         |                                       |



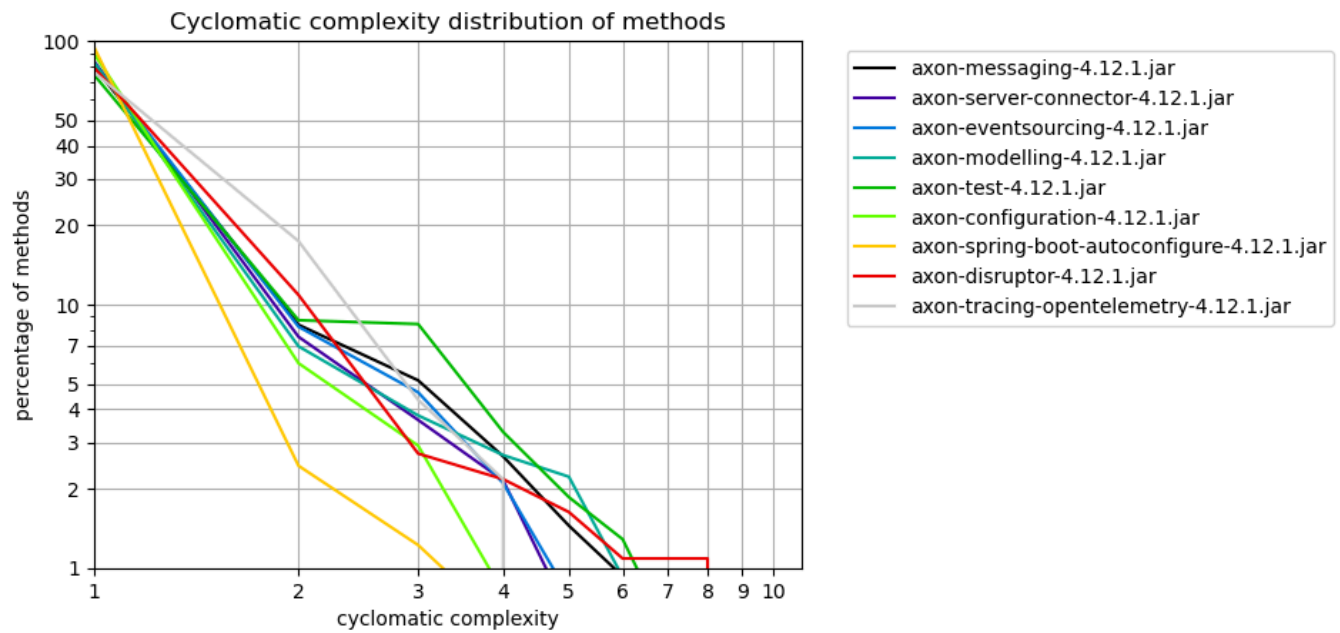
Table 2b - Cyclomatic method complexity distribution (normalized)

The table shown here only includes the first 40 rows which typically represents the most significant entries. Have a look below to find out which packages and methods have the highest effective lines of code.

| artifactName         | axon-messaging-4.12.1.jar | axon-server-connector-4.12.1.jar | axon-eventsourcing-4.12.1.jar | axon-modelling-4.12.1.jar | axon-test-4.12.1.jar | axon-configuration-4.12.1.jar | axon-spring-boot-autoconfigure-4.12.1.jar | axon-disruptor-4.12.1.jar | axon-tracing-opentelemetry-4.12.1.jar |
|----------------------|---------------------------|----------------------------------|-------------------------------|---------------------------|----------------------|-------------------------------|-------------------------------------------|---------------------------|---------------------------------------|
| cyclomaticComplexity |                           |                                  |                               |                           |                      |                               |                                           |                           |                                       |
| 1                    | 80.110307                 | 83.844242                        | 82.574431                     | 82.886216                 | 74.428571            | 89.482201                     | 94.908350                                 | 79.347826                 | 76.08695                              |
| 2                    | 8.376422                  | 7.539354                         | 8.231173                      | 6.938020                  | 8.714286             | 5.987055                      | 2.443992                                  | 10.869565                 | 17.39130                              |
| 3                    | 5.153395                  | 3.645402                         | 4.640981                      | 3.792784                  | 8.428571             | 2.912621                      | 1.221996                                  | 2.717391                  | 4.34782                               |
| 4                    | 2.654257                  | 2.154101                         | 2.101576                      | 2.682701                  | 3.285714             | 0.809061                      | 0.610998                                  | 2.173913                  | 2.17391                               |
| 5                    | 1.447777                  | 0.662800                         | 0.788091                      | 2.220167                  | 1.857143             | 0.485437                      | 0.203666                                  | 1.630435                  | 0.00000                               |
| 6                    | 0.930714                  | 0.414250                         | 0.350263                      | 0.925069                  | 1.285714             | 0.000000                      | 0.203666                                  | 1.086957                  | 0.00000                               |
| 7                    | 0.396415                  | 0.579950                         | 0.612960                      | 0.185014                  | 0.571429             | 0.323625                      | 0.203666                                  | 1.086957                  | 0.00000                               |
| 8                    | 0.241296                  | 0.497100                         | 0.612960                      | 0.092507                  | 0.285714             | 0.000000                      | 0.000000                                  | 1.086957                  | 0.00000                               |
| 9                    | 0.155119                  | 0.331400                         | 0.000000                      | 0.185014                  | 0.285714             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 10                   | 0.103413                  | 0.165700                         | 0.000000                      | 0.000000                  | 0.142857             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 11                   | 0.172354                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.285714             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 12                   | 0.086177                  | 0.082850                         | 0.000000                      | 0.000000                  | 0.285714             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 13                   | 0.034471                  | 0.000000                         | 0.087566                      | 0.000000                  | 0.142857             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 14                   | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.203666                                  | 0.000000                  | 0.00000                               |
| 15                   | 0.017235                  | 0.082850                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 16                   | 0.000000                  | 0.000000                         | 0.000000                      | 0.092507                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 17                   | 0.034471                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 21                   | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 23                   | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 40                   | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |
| 41                   | 0.017235                  | 0.000000                         | 0.000000                      | 0.000000                  | 0.000000             | 0.000000                      | 0.000000                                  | 0.000000                  | 0.00000                               |

Table 2b Chart 1 - Cyclomatic method complexity distribution (normalized)

<Figure size 640x480 with 0 Axes>



## Table 2c - Top 30 packages with highest cyclomatic complexity

The following table shows the top 30 packages with the highest cyclomatic complexity. The whole table can be found in the CSV report

`Effective_lines_of_method_code_per_package` .

|    | artifactName                          | fullPackageName                                   | complexityInPackage | methodCount | maxComplexity |                           |
|----|---------------------------------------|---------------------------------------------------|---------------------|-------------|---------------|---------------------------|
| 0  | axon-messaging-4.12.1                 | org.axonframework.eventhandling                   | 1276                | 838         | 21            |                           |
| 1  | axon-configuration-4.12.1             | org.axonframework.config                          | 730                 | 618         | 7             |                           |
| 4  | axon-messaging-4.12.1                 | org.axonframework.queryhandling                   | 484                 | 388         | 11            |                           |
| 2  | axon-messaging-4.12.1                 | org.axonframework.eventhandling.pooled            | 472                 | 350         | 23            |                           |
| 7  | axon-modelling-4.12.1                 | org.axonframework.modelling.command               | 458                 | 336         | 9             |                           |
| 5  | axon-server-connector-4.12.1          | org.axonframework.axonserver.connector.event.axon | 453                 | 283         | 15            |                           |
| 3  | axon-test-4.12.1                      | org.axonframework.test.aggregate                  | 438                 | 251         | 13            |                           |
| 8  | axon-server-connector-4.12.1          | org.axonframework.axonserver.connector            | 419                 | 335         | 12            |                           |
| 10 | axon-eventsourcing-4.12.1             | org.axonframework.eventsourcing.eventstore        | 397                 | 264         | 13            |                           |
| 12 | axon-messaging-4.12.1                 | org.axonframework.messaging.annotation            | 385                 | 238         | 14            |                           |
| 13 | axon-modelling-4.12.1                 | org.axonframework.modelling.command.inspection    | 336                 | 215         | 9             |                           |
| 14 | axon-eventsourcing-4.12.1             | org.axonframework.eventsourcing                   | 325                 | 251         | 8             |                           |
| 6  | axon-messaging-4.12.1                 | org.axonframework.eventhandling.deadletter.jdbc   | 304                 | 249         | 12            |                           |
| 20 | axon-messaging-4.12.1                 | org.axonframework.common                          | 301                 | 144         | 9             | getExactDirectSuperTypesO |
| 9  | axon-server-connector-4.12.1          | org.axonframework.axonserver.connector.query      | 295                 | 219         | 9             |                           |
| 11 | axon-spring-boot-autoconfigure-4.12.1 | org.axonframework.springboot.autoconfig           | 292                 | 249         | 14            |                           |
| 16 | axon-eventsourcing-4.12.1             | org.axonframework.eventsourcing.eventstore.leg... | 292                 | 187         | 8             |                           |
| 18 | axon-messaging-4.12.1                 | org.axonframework.serialization                   | 290                 | 184         | 7             |                           |
| 22 | axon-modelling-4.12.1                 | org.axonframework.modelling.saga                  | 277                 | 194         | 6             |                           |
| 15 | axon-disruptor-4.12.1                 | org.axonframework.disruptor.commandhandling       | 274                 | 184         | 8             |                           |
| 17 | axon-eventsourcing-4.12.1             | org.axonframework.eventsourcing.eventstore.jdbc   | 273                 | 236         | 7             | lambda fε                 |
| 23 | axon-messaging-4.12.1                 | org.axonframework.commandhandling.gateway         | 249                 | 174         | 12            |                           |
| 24 | axon-messaging-4.12.1                 | org.axonframework.commandhandling.distributed     | 243                 | 175         | 12            |                           |
| 21 | axon-test-4.12.1                      | org.axonframework.test.saga                       | 236                 | 169         | 9             |                           |
| 19 | axon-messaging-4.12.1                 | org.axonframework.eventhandling.deadletter.jpa    | 212                 | 132         | 15            |                           |
| 32 | axon-messaging-4.12.1                 | org.axonframework.messaging                       | 209                 | 156         | 4             |                           |

|    | artifactName                          | fullPackageName                        | complexityInPackage | methodCount | maxComplexity |
|----|---------------------------------------|----------------------------------------|---------------------|-------------|---------------|
| 29 | axon-messaging-4.12.1                 | org.axonframework.messaging.unitofwork | 206                 | 129         | 11            |
| 27 | axon-messaging-4.12.1                 | org.axonframework.commandhandling      | 201                 | 169         | 10            |
| 31 | axon-test-4.12.1                      | org.axonframework.test.matchers        | 191                 | 108         | 8             |
| 33 | axon-spring-boot-autoconfigure-4.12.1 | org.axonframework.springboot           | 191                 | 186         | 6             |

## Table 2d - Top 30 methods with highest cyclomatic complexity

The following table shows the top 30 packages containing the methods with the highest cyclomatic complexity. The whole table can be found in the CSV report

Effective\_lines\_of\_method\_code\_per\_package .

| index | artifactName | fullPackageName                                                                   | maxComplexityType                                       | maxComplexityMe       |
|-------|--------------|-----------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------|
| 0     | 61           | axon-messaging-4.12.1<br>org.axonframework.updates.detection                      | TestEnvironmentDetector                                 | isTest                |
| 1     | 66           | axon-messaging-4.12.1<br>org.axonframework.eventhandling.scheduling.job...        | JobRunrEventScheduler                                   | <i>deserializeLan</i> |
| 2     | 2            | axon-messaging-4.12.1<br>org.axonframework.eventhandling.pooled                   | Coordinator\$CoordinationTask                           |                       |
| 3     | 0            | axon-messaging-4.12.1<br>org.axonframework.eventhandling                          | TrackingEventProcessor                                  | processE              |
| 4     | 50           | axon-messaging-4.12.1<br>org.axonframework.updates.api                            | UpdateCheckResponse                                     | fromRec               |
| 5     | 40           | axon-modelling-4.12.1<br>org.axonframework.modelling.saga.repository              | AssociationValueMap\$AssociationValueComparator         | corr                  |
| 6     | 5            | axon-server-connector-4.12.1<br>org.axonframework.axonserver.connector.event.axon | PersistentStreamConnection <i>SegmentConnectionP...</i> | readMess              |
| 7     | 19           | axon-messaging-4.12.1<br>org.axonframework.eventhandling.deadletter.jp            | DeadLetterEventEntry                                    | ex                    |
| 8     | 11           | axon-spring-boot-autoconfigure-4.12.1<br>org.axonframework.springboot.autoconfig  | AxonAutoConfiguration                                   | buildSeria            |
| 9     | 12           | axon-messaging-4.12.1<br>org.axonframework.messaging.annotation                   | AnnotatedMessageHandlingMember                          | ha                    |
| 10    | 58           | axon-messaging-4.12.1<br>org.axonframework.deadline.jobrunr                       | JobRunrDeadlineManager                                  | <i>deserializeLan</i> |
| 11    | 67           | axon-messaging-4.12.1<br>org.axonframework.commandhandling.distributed....        | CommandNameFilter                                       | <i>deserializeLan</i> |
| 12    | 10           | axon-eventsourcing-4.12.1<br>org.axonframework.eventsourcing.eventstore           | ConcatenatingDomainEventStream                          | has                   |
| 13    | 3            | axon-test-4.12.1<br>org.axonframework.test.aggregate                              | AggregateTestFixture                                    | ensureValuest         |
| 14    | 23           | axon-messaging-4.12.1<br>org.axonframework.commandhandling.gateway                | CommandGatewayFactory                                   | createGat             |
| 15    | 6            | axon-messaging-4.12.1<br>org.axonframework.eventhandling.deadletter.jdbc          | JdbcDeadLetter                                          | ex                    |
| 16    | 48           | axon-messaging-4.12.1<br>org.axonframework.common.jdbc                            | ConnectionWrapperFactory                                | lambdaW               |
| 17    | 47           | axon-test-4.12.1<br>org.axonframework.test.server                                 | AxonServerContainer                                     | ex                    |
| 18    | 24           | axon-messaging-4.12.1<br>org.axonframework.commandhandling.distributed            | ReplyMessage                                            | ex                    |
| 19    | 8            | axon-server-connector-4.12.1<br>org.axonframework.axonserver.connector            | AxonServerConnectionManager\$Builder                    |                       |
| 20    | 34           | axon-messaging-4.12.1<br>org.axonframework.deadline.dbscheduler                   | DbSchedulerBinaryDeadlineDetails                        | ex                    |
| 21    | 4            | axon-messaging-4.12.1<br>org.axonframework.queryhandling                          | SimpleQueryBus                                          | doC                   |
| 22    | 26           | axon-messaging-4.12.1<br>org.axonframework.eventhandling.deadletter.leg...        | JpaDeadLetter                                           | ex                    |
| 23    | 29           | axon-messaging-4.12.1<br>org.axonframework.messaging.unitofwork                   | BatchingUnitOfWork                                      | executeWithR          |
| 24    | 43           | axon-messaging-4.12.1<br>org.axonframework.deadline.quartz                        | DeadlineJob                                             | exe                   |
| 25    | 36           | axon-messaging-4.12.1<br>org.axonframework.messaging.deadletter                   | GenericDeadLetter                                       | ex                    |

| index | artifactName          | fullPackageName                           | maxComplexityType                                  | maxComplexityMe |
|-------|-----------------------|-------------------------------------------|----------------------------------------------------|-----------------|
| 26    | axon-messaging-4.12.1 | org.axonframework.common.lock             | PessimisticLockFactory\$DisposableLock             |                 |
| 27    | axon-messaging-4.12.1 | org.axonframework.updates                 | UpdateCheckerHttpClient                            | sendRec         |
| 28    | axon-messaging-4.12.1 | org.axonframework.commandhandling         | MethodCommandHandlerDefinition\$MethodCommandMe... | <               |
| 29    | axon-messaging-4.12.1 | org.axonframework.messaging.responsetypes | MultipleInstancesResponseType                      | col             |